

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

An Implemented Blockchain-Based Framework for Healthcare Data Sharing

submitted in partial fulfilment of the requirements for the award of degree of

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In

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled “**An Implemented Blockchain-Based Framework for Healthcare Data Sharing**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Mrs. Swati Goel**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ABSTRACT

Health care is one of the largest generators of personal, private, and sensitive data in the world. With this large volume of sensitive patient information being exchanged securely across numerous entities such as hospitals, labs, insurance companies, and most importantly the patients themselves, centralized systems used to protect patient information have many shortcomings including data breaches, limited interoperability, lack of transparency in who can see what, and the patient's lack of control regarding their Personal Health Record (PHR).

As a result, there is increased interest in using blockchain — a distributed ledger system that is immutable and allows for peer-to-peer transactions without needing a central authority. This project provides a review of existing healthcare blockchain data sharing models, identifies their shortcomings, and proposes a patient-centric framework using smart contracts for access control and off-chain encrypted storage for scalable data management.

The proposed model increases data integrity, provides an auditable trail, empowers the patient to directly manage their medical record, and decreases dependency on centralized authorities. An experimental analysis compares the performance of the proposed model against other similar models to show its viability for actual real-world healthcare deployment.

Keywords — *Blockchain, Healthcare Data Sharing, Smart Contracts, Patient-Centric Security, Decentralized Systems, Electronic Health Records, Access Control.*

1. INTRODUCTION

Health care is one of the largest generators of personal, private, and sensitive data in the world. With this large volume of sensitive patient information being exchanged across numerous entities such as hospitals, labs, insurance companies, and the patients themselves, centralized systems have many shortcomings: data breaches, limited interoperability, lack of transparency, and patients' lack of control over their Personal Health Records (PHR). As a result, there is increased interest in blockchain technology, which is immutable and allows for peer-to-peer transactions without needing a central authority.

This paper makes the following contributions: (1) a survey of current blockchain-based healthcare data sharing approaches published between 2017 and 2024; (2) a conceptual framework for patient-centric blockchain-based data sharing using off-chain storage with on-chain access control; (3) a smart contract design for permission management enabling granular, auditable, and revocable access to medical records; and (4) a comparative evaluation demonstrating the practical advantages of the proposed framework over existing solutions.

A. Blockchain Technology

The term blockchain refers to an emerging technology enabling multiple parties to access a digital ledger through distributed systems. Each block contains a unique hash generated from the previous block, a timestamp, block ID, and transaction data. When a new block is created, every node updates its ledger copy, making unauthorized modifications virtually impossible.

Key characteristics of blockchain as it relates to healthcare:

- Decentralized: No central authority controls the ledger; every node stores a copy, creating redundancy and removing single points of failure.
- Immutable: Once placed on the blockchain, data cannot be changed without agreement from the majority of the network, creating a tamper-evident audit trail.
- Transparent: Every participant has visibility to all transactions, creating accountability and traceability.
- Smart Contracts: Self-executing programs stored on the blockchain that trigger predefined actions once specific conditions are met.
- Cryptographic Security: Every transaction is executed using private keys, validating sender identity and preventing repudiation.

B. Types of Blockchain Networks Relevant to Healthcare

- Public blockchain (Ethereum, Bitcoin): Open to everyone and completely decentralized but faces scaling and cost issues, making it unsuitable for high-volume healthcare data.
- Private blockchain (Hyperledger Fabric): Accessible only to permitted users; higher processing speed and better data security for enterprise-level healthcare environments.
- Consortium blockchain: Combines aspects of public and private blockchains, providing decentralization while maintaining confidentiality for healthcare applications.

C. Electronic Health Records (EHR)

Electronic Health Records are digital representations of a patient's complete history — diagnoses, medications, treatments, immunizations, allergies, lab tests, and imaging. Problems with current EHR systems include inconsistent standards, vendor lock-in, and limited access control. Blockchain provides an infrastructure layer that separates access control from data storage, resolving all four key issues.

2. RELATED WORK

A. Early Blockchain Healthcare Research

The systematic investigation into using blockchain technology in managing healthcare data began with Azaria et al.'s MedRec system, which demonstrated the feasibility of using Ethereum smart contracts to control permissioned access to patients' electronic health records — establishing the fundamental concept of separating access control logic (on-chain) from data storage (off-chain). Xia et al. proposed MeDShare, designed for cloud service providers to share medical data using blockchain, introducing provenance tracking to monitor data origins and access history.

B. Privacy-Protective Solutions

Dubovitskaya et al. developed a system for managing cancer patient data combining blockchain with cloud computing, using symmetric key cryptography and role-based access control to maintain patient data confidentiality across organizations. Bhattacharya et al. proposed zero-knowledge proof protocols, enabling users to prove possession of valid health credentials without revealing underlying details of those credentials.

C. Interoperability-Focused Systems

Gordon and Catalini discussed the potential role of blockchain in improving healthcare interoperability, suggesting standardized smart contract interfaces alongside HL7 FHIR data formats. Liang et al. created a mobile health data sharing system using lightweight cryptographic protocols suitable for resource-limited patient devices, with deployment results from a real-world clinical setting.

D. Access Control Mechanisms

Zhang et al. suggested using attribute-based encryption with smart contracts to create detailed access policies, allowing patients to decide which parts of their records are available to specific providers. Roehrs et al. introduced OmniPHR, a decentralized personal health record model allowing patients to gather and control records from various sources into a unified patient-controlled view.

3. LIMITATIONS OF EXISTING APPROACHES

A. Scalability Constraints

Public blockchain networks like Ethereum handle a limited number of transactions per second, making them unsuitable for high-volume healthcare settings where millions of access events can happen daily across large hospital networks. Even private blockchain solutions encounter throughput issues when applied at a national level.

B. Privacy and Confidentiality Gaps

Some proposed systems keep medical data directly on the blockchain, making it accessible to all network participants. While encryption helps reduce risks, long-term exposure of health data on immutable ledgers remains a concern as encryption standards evolve. Key management — ensuring patients can access their encrypted data after losing keys or devices — is often not effectively addressed.

C. Incomplete Patient Control

Many current frameworks claim patient-centered designs but still implement access control policies driven mainly by institutions. Patients frequently have to navigate complex interfaces to manage permissions, creating usability issues that hinder the patient empowerment goals of these systems.

D. Integration Complexity

Integrating blockchain technology with current EHR systems is challenging from both technological and organizational perspectives. Most proposals consider only newly installed EHR systems, ignoring the many hospitals that operate with legacy IT infrastructure that cannot be replaced immediately.

E. Lack of Emergency Access Policies

Current processes are inadequate for emergencies where the patient is unable to grant access. In the absence of emergency access policies, there is a risk of putting patients in harm's way when urgent medical decisions require record access.

4. RESEARCH METHODOLOGY

A. Research Design

This study utilizes a Design Science Research (DSR) paradigm, best suited to projects involving the development and assessment of an artifact — in this case, a blockchain-enabled framework for healthcare data exchange. The DSR process consists of three stages: (1) identification and motivation based on a literature review; (2) design and construction of the artifact as a conceptual model with smart contract coding; and (3) evaluation against performance and security criteria.

B. Literature Review Protocol

A systematic literature review was performed using IEEE Xplore, ACM Digital Library, Google Scholar, and PubMed with keywords including “blockchain and healthcare,” “smart contract for medical data,” and “decentralized EHRs.” Papers were included if they offered or critiqued a blockchain architecture for managing healthcare data, were published from 2017 to 2024, and were peer-reviewed conference papers or journal articles. Over 60 papers were identified; 8 primary architectures were selected for detailed consideration based on architectural significance and citation popularity.

C. Framework Design Methodology

The framework was designed following a requirement-driven strategy. The five limitations identified in Section 3 were transformed into design requirements: scalability exceeding 50 TPS; no raw health information stored on-chain; patient-only access grant authority; a compatible API for existing EHR integration; and an emergency access mechanism for unconscious patients. Each architectural decision was justified by a corresponding requirement, and alternatives such as purely on-chain storage were assessed and rejected due to privacy and scalability issues.

D. Smart Contract Design and Implementation

The smart contract was developed following OpenZeppelin guidelines for secure Ethereum smart contracts through four steps: (1) functional specification defining preconditions and postconditions for each function; (2) threat-driven design addressing reentrancy attacks, integer overflows, unsafe self-destruct, and front-running; (3) gas optimization via struct packing, mapping data structures for $O(1)$ lookups, and batched access grant creation; and (4) simulation testing on a local test Ethereum network.

E. Performance Evaluation Methodology

Performance evaluation used a comparative simulation approach modeling a regional hospital network with approximately 500 simultaneous users performing access grant, access request, and record retrieval operations. Transaction arrival followed a Poisson probability distribution. Measures included throughput

(TPS), end-to-end access latency, patient access control quality, and privacy level. Results from MedRec and MeDShare literature were used as calibration measures. Note that results reflect simulation, not live deployment.

F. Security Analysis Methodology

Security assessment involved threat modeling based on the STRIDE framework (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege). Threats were categorized within each framework component and assessed against adopted countermeasures, including private key authentication, on-chain hash verification for off-chain data, append-only audit logs, AES-256 encryption, decentralized replication, and owner-check functions in smart contracts.

5. PROPOSED FRAMEWORK

A. Framework Overview

The framework uses a three-layered architecture segregating responsibilities for data storage, access control policy, and human-computer interaction. Four key principles guide the design: (i) minimizing data on-chain; (ii) patient-only control over permissions; (iii) compatibility with existing EHR infrastructure; and (iv) a reliable emergency access mechanism.

B. Architectural Layers

1) Storage Layer: Patient medical records are stored off-chain using AES-256 encryption via IPFS (InterPlanetary File System). The unique encryption key is further encrypted and stored on the blockchain. A content hash of the stored information is placed on the blockchain for integrity checks — any alteration of stored data would be reflected in the hash value.

2) Access Control Layer: Access control is carried out using Ethereum-based smart contracts on a consortium blockchain among participating healthcare organizations. The smart contract maintains an Access Control List (ACL) for each patient detailing which providers have valid access, the level of access (read-only or read-write), record type, and validity period. The decryption key is returned only after the smart contract confirms both identity and a legitimate access grant.

3) Application Layer: Web and mobile applications enable patients to control and monitor permissions. Patients can view all access attempts, grant limited access to selected professionals, revoke access at any time, and designate emergency contacts with limited keys. Healthcare providers interact via APIs integrated into current EHR systems using FHIR R4-based interfaces.

C. Emergency Access Protocol

An access system based on Shamir's Secret Sharing technique distributes the master key across several trustworthy entities (for example, two family members and one institutional key escrow). In an emergency, any two of these three key shares can recreate the master key, enabling access without the patient's active permission while maintaining a full blockchain audit trail.

6. SMART CONTRACT DESIGN

A. Contract Design

The access control smart contract is coded in Solidity and executed on a permissioned Ethereum blockchain.

Core data structures and functions include:

- **PatientRegistry**: Maps patient identifiers to their encrypted record hash and access control metadata.
- **AccessGrant**: A structure comprising provider address, scope, expiry timestamp, and access status.
- **grantAccess(address provider, bytes32 scope, uint256 duration)**: Invoked by patients to grant access to a provider.
- **revokeAccess(address provider)**: Invoked by patients to immediately revoke access from a provider.
- **requestAccess(address patient, bytes32 scope)**: Invoked by providers, triggering an event for key exchange once verified.
- **auditLog**: An immutable on-chain collection of all access events.

B. Event-Driven Key Delivery

To prevent storing decryption keys on-chain, the smart contract implements an event-driven approach. Upon successful permission verification, the contract fires an “AccessGranted” event. An off-chain oracle service subscribed to this event securely forwards the decryption key to the authorized user via encrypted channels, keeping private keys entirely off-chain.

C. Gas Optimization

Optimization measures include storage packing of struct variables, using mapping data structures instead of arrays for constant-time $O(1)$ lookups on the PatientRegistry and auditLog, and batched access grant creation to save on base transaction gas fees. Each access grant transaction costs approximately 45,000 gas.

7. SECURITY ANALYSIS

A. Threat Model

The threat model considers three adversary types: external adversaries seeking unauthorized access to patient information; insiders in healthcare organizations attempting to access records outside their authorization level; and honest-but-curious participants attempting to infer information from metadata observed during protocol use.

B. Security Properties

- Confidentiality: Patient records are AES-256 encrypted. Records stored in distributed storage cannot be differentiated from random data without the patient's decryption key.
- Integrity: Content hash values ensure no unauthorized tampering with stored records. Any alteration is immediately detectable.
- Authentication: Blockchain transaction signatures ensure only authorized users can perform grant or revoke operations.
- Non-repudiation: An append-only audit log on the blockchain ensures that the full history of all accesses can never be repudiated.
- Availability: IPFS-based distributed storage eliminates single points of failure. Blockchain replication guarantees access control layer availability even when individual nodes fail.

C. Attack Resistance Analysis

- 51% Attacks: For a consortium blockchain with known, trusted member organizations, a 51% attack would require compromising more than half of all validator nodes — highly impractical for a well-governed healthcare consortium.
- Access Control Contract Security: The contract follows secure coding best practices to avoid reentrancy attacks, integer overflows, and unguarded self-destruct functionality. Formal verification is recommended for production deployment.
- Key Compromise: If a patient's private key is compromised, an attacker can modify access permissions but cannot decrypt stored health data without also compromising the separate cloud encryption keys.

8. PERFORMANCE EVALUATION

A. Evaluation Methodology

The new system was assessed using simulations and comparisons against three existing systems: MedRec [1], MeDShare [2], and a centralized EHR baseline. Performance criteria include transaction rate (TPS), average access grant latency, patient control level, auditability, scalability, privacy level, interoperability, and emergency access support.

B. Comparative Results

TABLE I: Comparative Evaluation of Blockchain Healthcare Data Sharing Frameworks

Metric	MedRec [1]	MeDShare [2]	Centralized EHR	Proposed Framework
Throughput (TPS)	15–20	18–25	500+	80–120
Access Latency	3–5 sec	2–4 sec	<1 sec	1–2 sec
Patient Control	Partial	Limited	None	Full
Auditability	Partial	Full	Limited	Full
Scalability	Low	Medium	High	High
Privacy Level	Medium	Medium	Low	High
Interoperability	Low	Low	Medium	High
Emergency Access	No	No	Yes	Yes

C. Discussion

The throughput of 80–120 TPS achieved by the proposed framework is significantly higher than existing healthcare blockchains due to its permissioned consortium architecture that avoids proof-of-work consensus. Although lower than centralized systems, 80–120 TPS is sufficient for most healthcare networks and can be further improved through horizontal scaling and sharding.

An access latency of 1–2 seconds is acceptable for non-emergency healthcare processes and considerably better than other blockchain proposals. Full patient control, complete auditability, strong privacy protection, and emergency access together make the proposed solution uniquely comprehensive among all analyzed systems.

9. IMPLEMENTATION CONSIDERATIONS

A. Technology Stack

The reference implementation uses: Hyperledger Besu for permissioned Ethereum-compatible blockchain infrastructure; IPFS for off-chain distributed storage; Solidity for writing smart contracts; OpenZeppelin for secure contract libraries; and React.js along with Web3.js for developing the patient-facing application interface.

B. Deployment Architecture

Deployment in actual healthcare environments requires creating a consortium blockchain where each participating entity runs one or more validator nodes. Governance policies regulate how validators join and leave the chain, how protocol updates happen, and what occurs during network emergencies. The system integrates with existing EHR systems through a middleware component that exposes FHIR R4-based APIs translating between blockchain access control commands and current clinical workflow systems.

C. Regulatory Compliance

Healthcare data management is subject to stringent regulatory requirements including HIPAA in the United States, GDPR in Europe, and the Personal Data Protection Bill in India. The proposed framework's patient consent mechanism, audit logging, data minimization approach, and right-to-access support align with these regulatory frameworks. Legal analysis specific to the deployment jurisdiction is strongly recommended prior to production deployment.

10. CONCLUSION AND FUTURE WORK

This paper has provided a comprehensive review of methods for sharing healthcare data using blockchain and proposes a patient-focused framework addressing the key limitations of current solutions. The framework combines off-chain encrypted storage with on-chain smart contract-based access control, striking a practical balance among security, privacy, scalability, and usability. The research demonstrates that the framework provides patients with better control, stronger privacy, and improved auditability compared to both existing blockchain and centralized alternatives, while delivering adequate performance for real-world use.

Future work will focus on: (1) formally verifying the smart contract using tools like Certora Prover; (2) developing and user-testing the patient-facing mobile application with actual patient groups; (3) pilot deployment among a group of regional healthcare providers; (4) integrating with national health ID systems for improved patient authentication; and (5) exploring layer-2 solutions to further increase transaction speed for large-scale national healthcare networks.

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